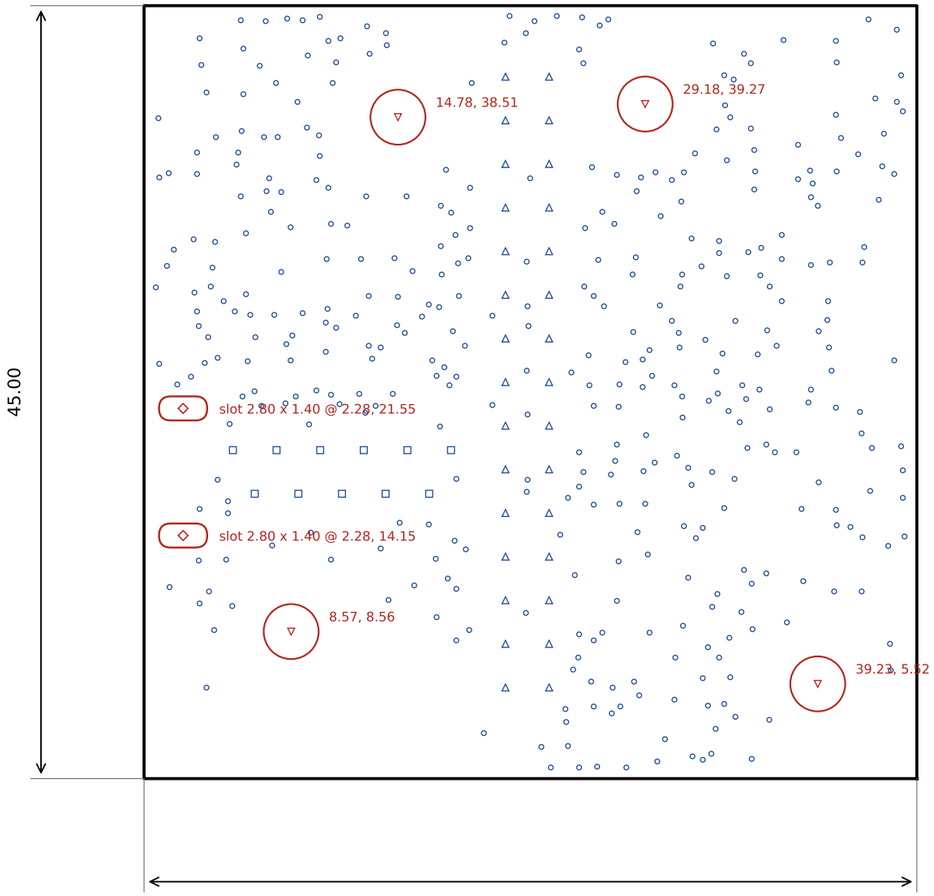


Datum: lower-left corner of the outline (0, 0). Dimensions in mm, outline tolerance +/-0.2 mm.
Drawn from the Excellon files: every drilled hole is shown with its symbol; holes >= 2 mm also to scale.

BOOST CONTROL CARD - FABRICATION DRAWING

BOOST_control_RC2, 2026-10-13. Files: BOOST_control_RC2_gerbers.zip. Units: mm.

FABRICATION NOTES

1. **ALL VIAS EPOXY FILLED AND COPPER CAPPED (resin plugged, plated over; IPC-4761 Type VII). 188 of the 398 vias sit in SMD pads. Via drills 0.30 mm (JLCPCB filled-via range 0.2-0.5 mm). Order option "Via Covering: Epoxy Filled & Capped".**
2. Material FR-4, Tg 155 C or better (JLCPCB "FR4 TG155"). 6 copper layers. Finished thickness 1.6 mm +/-10 %.
3. Copper: outer layers 1 oz (35 um) finished; the four inner layers 0.5 oz (order option "Inner copper weight 0.5 oz", the fab default for 6 layers).
4. Stackup: the fab's standard 6-layer 1.6 mm build for 1 oz outer / 0.5 oz inner copper (no stackup specified), as in the table. No impedance control.
5. Surface finish ENIG. Solder mask green LPI both sides. Silkscreen white both sides.
6. Minimum track / space: 0.15 / 0.20 mm on every layer. Minimum drill 0.30 mm (vias); smallest plated pad hole 1.00 mm.
7. Minimum via annular ring 0.10 mm; minimum plated pad annular ring 0.33 mm. Via-to-via hole gap >= 0.30 mm; filled via to any pad hole >= 0.59 mm.
8. Plated hole tolerance +0.13 / -0.08 mm. J11 2x15 header holes are drilled 1.05 mm for Samtec 0.64 mm square posts.
9. Non-plated holes: 4 x 3.20 mm (standoffs) and 2 slots 2.80 x 1.40 mm (J9 cable tie), routed; the slots are G85 routes in the NPTH drill file.
10. Outline on the Edge.Cuts (GM1) layer, tolerance +/-0.2 mm.
11. Electrical test: 100 % flying probe; IPC-D-356 netlist BOOST_control_RC2.ipc356 supplied.
12. No order number or other marking to be added ("Remove Mark").
13. Assembly panel: standard PCBA needs at least 70 x 70 mm; order "Panel by JLCPCB" (edge rails) for this 45 x 45 mm card.

STACKUP (from the board file; copper + dielectric 1.547 mm, + mask)

F.Mask	solder mask		0.010
F.Cu	copper, 1 oz	signal / parts	0.035
	prepreg 3313, er 4.1		0.0994
In1.Cu	copper, 0.5 oz	signal / GND fill	0.0152
	core, er 4.6		0.550
In2.Cu	copper, 0.5 oz	signal / 5V pour	0.0152
	prepreg 2116, er 4.16		0.1164
In3.Cu	copper, 0.5 oz	signal / GND fill	0.0152
	core, er 4.6		0.550
In4.Cu	copper, 0.5 oz	signal / GND fill	0.0152
	prepreg 3313, er 4.1		0.0994
B.Cu	copper, 1 oz	signal / parts	0.035
B.Mask	solder mask		0.010

DRILL TABLE

sym	drill mm	type	count	used for
○	0.30	PTH	398	vias (filled, capped)
□	1.00	PTH	11	J9 wire pads
△	1.05	PTH	30	2x15 header (J11)
◇	1.40	NPTH slot	2	J9 tie slots 2.8 x 1.4
▽	3.20	NPTH	4	standoffs